# IIoT and Cloud Computing (CS5005)

Lecture – 26

SLA

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# What is Service Level Agreement?

- A formal contract between a Service Provider (SP) and a Service Consumer (SC)
- SLA: foundation of the consumer's trust in the provider
- Purpose : to define a formal basis for performance and availability the SP guarantees to deliver
- SLA contains Service Level Objectives (SLOs), SLI, Penalty Model, Measurement Window
  - Objectively measurable conditions for the service
  - SLO: basis of selection of cloud provider – What cloud has promised?
- SLI (Service Level Indicator) – What we measure?

# SLA Contents

- A set of services which the provider will deliver
- A complete, specific definition of each service
- The responsibilities of the provider and the consumer
- A set of metrics to measure whether the provider is offering the services as guaranteed
- An auditing mechanism to monitor the services (by a 3rd Party)
- The remedies available (payback) to the consumer and the provider if the terms are not satisfied
- How the SLA will change over time

# Types of SLA

- Present market place features two types of SLAs :
  - Off-the-shelf SLA or non-negotiable SLA or Direct SLA
    - Non-conducive for mission-critical data or applications
    - Provider creates the SLA template and define all criteria like contract period, billing, response time, availability, etc.
    - *Followed by the present day state-of-the-art clouds.*
  - Negotiable SLA
    - Negotiation via external agent
    - Negotiation via multiple external agents
    - If SC (service consumer) is a large organization

# Service Level Objectives (SLOs)

- Objectively measurable conditions for the service
- Encompasses multiple QoS parameters viz. availability, throughput, response time, billing, penalties.
- Example :
  - "***Availability*** of a service X is 99.9%"
  - "***Response time*** of a database query Q is between 3 to 5 seconds"
  - "***Throughput*** of a server S at peak load time is 0.875"
- SLA = <A, L, T, R>
  - Availability
  - Latency
  - Throughput
  - Reliability

# Service Level Management

- Monitoring and measuring performance of services based on SLOs
- Provider perspective :
  - Make decisions based on business objectives and technical realties
- Consumer perspective :
  - Decisions about how to use cloud services